

Inequalities

Some collect stamps, some collect coins, some collect inequalities. (Anonymous).

Inequalities are more important than equalities, as things are more likely not equal than equal. (Anonymous).

Remark: The most basic inequalities are the triangle inequality in geometry and the AM-GM inequality.

(AM-GM-HM Inequality) For $a_1, a_2, \dots, a_n > 0$,

$$AM = \frac{a_1 + a_2 + \dots + a_n}{n} \geq GM = \sqrt[n]{a_1 a_2 \dots a_n} \geq HM = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}}.$$

Equalities hold if and only if all a_i 's are equal.

For the AM-GM inequality, the a_i 's may be relaxed to all $a_i \geq 0$.

(Cauchy-Schwarz Inequality) Let $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ be real numbers. Then

$$(a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2) \geq (a_1 b_1 + a_2 b_2 + \dots + a_n b_n)^2.$$

Equality holds if and only if $a_i b_j = a_j b_i$ for $1 \leq i, j \leq n$.

(Triangle Inequality) Let $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ be real numbers. We have

$$\sqrt{a_1^2 + a_2^2 + \dots + a_n^2} + \sqrt{b_1^2 + b_2^2 + \dots + b_n^2} \geq \sqrt{(a_1 + b_1)^2 + (a_2 + b_2)^2 + \dots + (a_n + b_n)^2}.$$

(Jensen Inequality) Let $f: [a, b] \rightarrow \mathbb{R}$ a convex function. (Example: $f(x) = x^2$).

Then for any $x_1, x_2, \dots, x_n \in [a, b]$ and any nonnegative reals $\omega_1, \omega_2, \dots, \omega_n$ with

$\omega_1 + \omega_2 + \dots + \omega_n = 1$, we have

$$\omega_1 f(x_1) + \omega_2 f(x_2) + \dots + \omega_n f(x_n) \geq f(\omega_1 x_1 + \omega_2 x_2 + \dots + \omega_n x_n).$$

If f is concave, the inequality is reversed.

(Weighted AM-GM). Let $\omega_1, \omega_2, \dots, \omega_n > 0$, with $\omega_1 + \omega_2 + \dots + \omega_n = 1$. For all $x_1, x_2, \dots, x_n > 0$, we have

$$\omega_1 x_1 + \omega_2 x_2 + \dots + \omega_n x_n \geq x_1^{\omega_1} x_2^{\omega_2} \dots x_n^{\omega_n}.$$

(Schur) Let x, y, z be nonnegative real numbers. For any $r > 0$, we have

$$\sum_{cyclic} x^r (x - y)(x - z) \geq 0.$$

(Power Mean) Let $x_1, x_2, \dots, x_n > 0$. The *power mean of order r* is defined by

$$M_{(x_1, x_2, \dots, x_n)}(0) = \sqrt[n]{x_1 x_2 \dots x_n}, M_{(x_1, x_2, \dots, x_n)}(r) = \left(\frac{x_1^r + x_2^r + \dots + x_n^r}{n} \right)^{\frac{1}{r}} (r \neq 0).$$

Then $M_{(x_1, x_2, \dots, x_n)} : \mathbb{R} \rightarrow \mathbb{R}$ is *continuous* and *monotone increasing*.

(Bernoulli) For all $r \geq 1$ and $x \geq -1$, we have

$$(1 + x)^r \geq 1 + rx.$$

(Symmetric Mean) For given arbitrary real numbers $x_1, x_2, \dots, x_n$, the coefficient of t^{n-i} in the polynomial $(t + x_1)(t + x_2) \dots (t + x_n)$ is called the i^{th} *elementary symmetric function* σ_i . This means that

$$(t + x_1)(t + x_2) \dots (t + x_n) = \sigma_0 t^n + \sigma_1 t^{n-1} + \dots + \sigma_{n-1} t + \sigma_n$$

For $1 \leq i \leq n$, the i^{th} *elementary symmetric function* S_i is defined by $S_i = \frac{\sigma_i}{\binom{n}{i}}$.

We have for positive x_i 's,

(Newton) $\frac{S_i}{S_{i+1}} \geq \frac{S_{i-1}}{S_i}.$

(Maclaurin) $S_i^{1/i} \geq S_{i+1}^{1/(i+1)}.$

(Rearrangement) Let $x_1 \geq x_2 \geq \dots \geq x_n$ and $y_1 \geq y_2 \geq \dots \geq y_n$ be real numbers. For any permutation σ of $\{1, 2, \dots, n\}$, we have

$$\sum_{i=1}^n x_i y_i \geq \sum_{i=1}^n x_i y_{\sigma(i)} \geq \sum_{i=1}^n x_i y_{n+1-i}.$$

(Chebyshev) Let $x_1 \geq x_2 \geq \dots \geq x_n$ and $y_1 \geq y_2 \geq \dots \geq y_n$ be real numbers. We have

$$\frac{x_1 y_1 + x_2 y_2 + \dots + x_n y_n}{n} \geq \left(\frac{x_1 + x_2 + \dots + x_n}{n} \right) \left(\frac{y_1 + y_2 + \dots + y_n}{n} \right).$$

(Holder) As generalized Cauchy. Let $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$ be positive real numbers. Suppose $p > 1, q > 1$, and $\frac{1}{p} + \frac{1}{q} = 1$. Then we have

$$\sum_{i=1}^n x_i y_i \leq \left(\sum_{i=1}^n x_i^p \right)^{1/p} \left(\sum_{i=1}^n y_i^q \right)^{1/q}.$$

More generally, let $x_{ij}, 1 \leq i \leq m, 1 \leq j \leq n$ be positive real numbers, also $\omega_1, \omega_2, \dots, \omega_n$ be positive real numbers satisfying $\omega_1 + \omega_2 + \dots + \omega_n = 1$. Then we have

$$\prod_{j=1}^n \left(\sum_{i=1}^m x_{ij} \right)^{\omega_j} \geq \sum_{i=1}^m \left(\prod_{j=1}^n x_{ij}^{\omega_j} \right).$$

(Minkowski) Let $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$ be positive real numbers and $p > 1$, then

$$\left(\sum_{i=1}^n x_i^p \right)^{1/p} + \left(\sum_{i=1}^n y_i^p \right)^{1/p} \geq \left(\sum_{i=1}^n (x_i + y_i)^p \right)^{1/p}.$$

(Majorization) Let $\vec{x} = (x_1, x_2, \dots, x_n)$ and $\vec{y} = (y_1, y_2, \dots, y_n)$ be two sequences of real numbers. $\vec{x}$ is said to *majorize* $\vec{y}$ ($\vec{x} \succ \vec{y}$) if

- (i) $x_1 \geq x_2 \geq \dots \geq x_n$ and $y_1 \geq y_2 \geq \dots \geq y_n$.
- (ii) $x_1 + x_2 + \dots + x_k \geq y_1 + y_2 + \dots + y_k$ for $1 \leq k \leq n-1$.
- (iii) $x_1 + x_2 + \dots + x_n = y_1 + y_2 + \dots + y_n$.

(Muirhead) Suppose that $(a_1, a_2, \dots, a_n) \succ (b_1, b_2, \dots, b_n)$ and $x_1, x_2, \dots, x_n$ are positive real numbers, then

$$\sum_{sym} x_1^{a_1} x_2^{a_2} \dots x_n^{a_n} \geq \sum_{sym} x_1^{b_1} x_2^{b_2} \dots x_n^{b_n},$$

where the symmetric is taken over all $n!$ permutation of $x_1, x_2, \dots, x_n$.

(Karamata Majorization Inequality) Let $f: [a, b] \rightarrow \mathbb{R}$ be a convex function. Suppose that $(x_1, x_2, \dots, x_n) \succ (y_1, y_2, \dots, y_n)$, with x_i 's and y_i 's belong to the interval $[a, b]$. Then

$$f(x_1) + f(x_2) + \dots + f(x_n) \geq f(y_1) + f(y_2) + \dots + f(y_n).$$

Remarks:

- (i) Avoid to use Muirhead if there are more than 3 variables.
- (ii) Do so only if the expression is symmetric, not cyclic.
- (iii) It is basically a form of weighted AM-GM mean.

(Example). Check:

- (i) $(0, 3, 0) \succ (1, 0, 2)$? Yes, 0, 3, 0 is the sequence 3, 0, 0.
- (ii) $(4, 0, 0, 0) \succ (2, 0, 2)$? No, different numbers of elements in the sequences.
- (iii) $(5, 0, -1) \succ (2, 2, 0)$? No, negative elements not allowed.
- (iv) $(2, 1, 1, 1) \succ (1, 1, 1, 1)$? No, the sums differ.
- (v) $(4, 1, 1, 1) \succ (3, 3, 1, 0)$? No, since $4+1 < 3+3$.

(Example) For positive x, y , show that (i) $x^2 + y^2 \geq 2xy$ and (ii) $x^3 + y^3 + z^3 \geq 3xyz$.
For (i), may apply Muirhead $(2, 0) \succ (1, 1)$. For (ii) $(3, 0, 0) \succ (1, 1, 1)$.

(Example) For positive x, y , show that $2(x^5 + y^5) \geq (x^2 + y^2)(x^3 + y^3)$.

$(5, 0) \succ (3, 2)$, hence $x^5 + y^5 \geq x^3y^2 + x^2y^3$. Together

$$\begin{aligned} 2(x^5 + y^5) &\geq (x^5 + y^5) + (x^3y^2 + x^2y^3) \\ &= (x^2 + y^2)(x^3 + y^3). \end{aligned}$$

(Example, BMO 2002). For positive x, y and z satisfying $x^2 + y^2 + z^2 = 1$. Show that

$$x^2yz + xy^2z + xyz^2 \leq \frac{1}{3}.$$

Note $(4, 0, 0) \succ (2, 1, 1)$ and $(2, 2, 0) \succ (2, 1, 1)$. Therefore

$$x^4 + y^4 + z^4 \geq x^2yz + xy^2z + xyz^2 \quad \text{and} \quad x^2y^2 + y^2z^2 + z^2x^2 \geq x^2yz + xy^2z + xyz^2.$$

Adding two times of the second expression and the first, get

$$(x^4 + y^4 + z^4) + 2(x^2y^2 + y^2z^2 + z^2x^2) \geq 3(x^2yz + xy^2z + xyz^2), \text{ or}$$

$$1 = (x^2 + y^2 + z^2)^2 \geq 3(x^2yz + xy^2z + xyz^2), \text{ hence the result.}$$

(Example, CHKMO2018) Given that a, b and c are positive real numbers such that

$$ab + bc + ca \geq 1, \text{ prove that } \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \geq \frac{\sqrt{3}}{abc}.$$

(First Solution)

Since

$$\begin{aligned} \frac{bc}{a} + \frac{ca}{b} &= c\left(\frac{b}{a} + \frac{a}{b}\right) \geq 2c, \\ \frac{ca}{b} + \frac{ab}{c} &= a\left(\frac{c}{b} + \frac{b}{c}\right) \geq 2a, \\ \frac{ab}{c} + \frac{bc}{a} &= b\left(\frac{a}{c} + \frac{c}{a}\right) \geq 2b. \end{aligned}$$

Adding the three inequalities, get

$$\frac{bc}{a} + \frac{ca}{b} + \frac{ab}{c} \geq a + b + c. \quad (1).$$

But

$$\begin{aligned} a + b + c &= \sqrt{(a^2 + b^2 + c^2) + 2(ab + bc + ca)} \\ &\geq \sqrt{3(ab + bc + ca)} \geq \sqrt{3}. \end{aligned} \quad (2).$$

From (1) and (3), we get $\frac{bc}{a} + \frac{ca}{b} + \frac{ab}{c} \geq \sqrt{3}$.

Hence $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \geq \frac{\sqrt{3}}{abc}$.

Remark: It is common to split a group of three into three possible pairs and consider them one by one.

(Alternate Solution). First $a^2 + b^2 + c^2 \geq ab + bc + ca$ by completing square, or Muirhead, or rearrangement inequality. Hence

$(a + b + c)^2 \geq 3(ab + bc + ca) \geq 3$, thus $a + b + c \geq \sqrt{3}$. Thus it suffices to show

$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \geq \frac{a + b + c}{abc}$, or equivalently $b^2c^2 + c^2a^2 + a^2b^2 \geq abc(a + b + c)$. This is immediate by Muirhead as $(2, 2, 0) \succ (2, 1, 1)$.

(Exercise) Let a, b, c be positive real numbers. Show that

(i) $4(a^3 + b^3) \geq (a + b)^3$.

(ii) $9(a^3 + b^3 + c^3) \geq (a + b + c)^3$.

(Example IMO2001) For positive real numbers a, b and c , show that

$$\frac{a}{\sqrt{a^2 + 8bc}} + \frac{b}{\sqrt{b^2 + 8ca}} + \frac{c}{\sqrt{c^2 + 8ab}} \geq 1.$$

Since $(a, b, c) \rightarrow (ka, kb, kc)$ will not change anything, we may assume $abc = \frac{1}{8}$,

get $\frac{a}{\sqrt{a^2 + 1/a}} + \frac{b}{\sqrt{b^2 + 1/b}} + \frac{c}{\sqrt{c^2 + 1/c}} \geq 1$. Let each of the three terms be x, y and z

respectively. Solve to get $a^3 = \frac{x^2}{1-x^2}, b^3 = \frac{y^2}{1-y^2}, c^3 = \frac{z^2}{1-z^2}$. Thus we have to prove

$x + y + z \geq 1$, assuming $abc = \frac{1}{8}$. By considering the contrapositive it suffices to

prove $abc < \frac{1}{8}$ by assuming $x + y + z < 1$. The squares have now gone and we need

to prove $\frac{x^2 y^2 z^2}{(1-x^2)(1-y^2)(1-z^2)} < \frac{1}{8^3}$. Now

$1-x^2 > (x+y+z)^2 - x^2 = y^2 + z^2 + xy + xy + yz + yz + zx + zx \geq 8x^{\frac{4}{8}}y^{\frac{6}{8}}z^{\frac{6}{8}}$. Likewise

for $1-y^2$ and $1-z^2$. put back into the equation to get the desired result.

(Using Jensen) We may assume $a+b+c=1$. As $x \mapsto \frac{1}{\sqrt{x}}$ is convex. By Jensen,

get

$$\begin{aligned} a \cdot \frac{1}{\sqrt{a^2+8bc}} + b \cdot \frac{1}{\sqrt{b^2+8ca}} + c \cdot \frac{1}{\sqrt{c^2+8ab}} &\geq \frac{1}{\sqrt{a(a^2+8bc)+b(b^2+8ca)+c(c^2+8ab)}} \\ &= \frac{1}{\sqrt{a^3+b^3+c^3+24abc}}. \end{aligned}$$

Thus it suffices to show

$$\frac{1}{\sqrt{a^3+b^3+c^3+24abc}} \geq 1, \text{ or equivalently } a^3+b^3+c^3+24abc \leq (a+b+c)^3. \text{ This}$$

can be done by AM-GM mean which gives

$$(a+b+c)^3 - (a^3+b^3+c^3) = 3(a^2b+b^2a+b^2c+c^2b+c^2a+a^2c) + 6abc \geq 24abc.$$

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